

Fascial form in yoga

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Introduction

Yoga is not about stretching (Chapter 9). Yoga is about fostering balance in the length and tension relationships in the body. This approach preserves one of the most valuable features of the fascial matrix: its elasticity. Stretching is but one aspect of promoting elasticity. We are, essentially, designed for storing the potential energy of recoil (Chapter 7). In practice, stretching is not always the way to do that.

Certain body types, or body histories, can mean that stretching for length is the last thing some people need. The kind of stretching that simply pulls on the tissues (Richards, 2012) can be

potentially harmful to the integrity of the whole. Yoga and fascia are both beautifully designed to honor everyone's ability to manage movement forces, as distinct from forced movements.

True elasticity is one of several fundamental principles of the fascial matrix profoundly enhanced by the appropriate practice of yoga. It incorporates a reciprocal relationship, such as the balance of tension and compression offers (Chapter 11). The growing understanding of the fascial tensional network makes sense of yoga, in all its potential for health and vitality. The optimization of tensional integrity of the tissues including the whole body, soft-tissue architecture can be depleted by efforts that over-stretch,



Figure 37.1 Tensional integrity or biotensegrity, is expressed as a whole, in the round

Forcing one part beyond its elastic limits (i.e. by stretching beyond "biotensegral range") can compromise another part. Whatever the pose, the fascial forms that we are made of animate (or are animated by) the whole body in the round. Thus, we take a global approach to overall integrity to enhance elastic recoil potential in individual ways for each individual – whatever the posture. The difference between these two images is that on the left Helen is inhaling, while on the right she is exhaling. There is a subtle, global effect. The tube is used as a metaphor, to imply global change without forcing stretch, facilitating an intimate balance through the relationship between breath and form.

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just as appropriate types of stretching can invigorate and strengthen it (see page 430).

The unifying experience

Yoga did not develop under the classical laws of Cartesian reductionist theory that have dominated Western anatomy, physiology, biomechanics and psychology for centuries. As an art and science, it was only ever based on the unifying experience of mind, body and being. Its ancient wisdom had no history of treating these aspects of “humans being” and “humans doing” separately.

Reducing “humans doing” (be it yoga or any movement modality) to functions and actions of muscles, bones, nerves and linear theories of biomechanical levers sits awkwardly with our fully animated and instinctive experience on the yoga mat. The properties of the fascia, as a ubiquitous sensory tensional network of tissues, throughout the rounded wholeness of our entire form, make perfect sense of it. In many ways it could be considered to perfectly express the biotensegrity principles of tension and compression (Chapter 11) united as volume; to incorporate the wholeness we express, whatever we are doing.

The yoga postures (asanas) in all their rich variety of shape and position, range and dynamics, are a means to explore balance and restore energy to those tissues, if we remain awake to their tensional integrity. Therein lies the containment of energy, the natural elasticity of compliance and our vitality. Even in Corpse Pose (*Shivasana*), the body rests as a pre-tensioned architecture (Van der Wal, 2009). We do not turn into an amoebic puddle on the mat when we are not moving, nor do we take the shape of the postures.

“A recognised characteristic of connective tissue is its impressive adaptability. When regularly put under increasing physiological strain, it changes its architectural properties to meet the increasing demand.”

Schleip (2011).

We have considerable power over the nature of the demand we put on the tissues. If we place no strain on them at all, it is the demand for inertia that effectively causes its own “lack of” strain patterns. This can lead to a requirement to tension the tensional network rather than stretch a relatively untensioned one. That is not to say that certain types of stretching are not valuable to all bodies. Consider instinctive stretching or pandiculation (Bertolucci, 2011). This can wake up sedentary tissues and relieve tight ones. However, it is distinct from the kind of stretching for which yoga sometimes has a reputation, one in which reaching for maximum length for its own sake can be considered a worthy goal to “achieve” a posture.

Some yoga practices are dedicated to long sequences, which also suit the development of elastic momentum. However, there is a caveat under the fasciological theme of tensional integrity (Chapter 11). If a dynamic series is repeated at very regular and frequent intervals all in one direction of strain, for example, flexion biased, it is likely to have a cumulative effect that can potentially encourage a particular strain pattern, which can cost elasticity in the longer term. However, if it is regularly counterbalanced with a series that is, for example, extension biased, then this pattern can be balanced and be more beneficial to the whole. The fascial matrix is a refined force transmission system

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(Langevin, 2006) and will respond according to the forces put through it (Schleip, 2011, see the above quotation).

It is possible to ensure an equally refined balance of forces to optimize the elastic capacity and recoil facility of the body (Chapter 11) without overstraining it in one particular direction. Once we understand the polarities involved and the definition of elasticity, we can accumulate the benefits of increased resilience, compliance and spring loading throughout our systems. It gives us moment-to-moment and movement-to-movement balance at naturally instinctive speed, appropriate to the individual and tissue type (Avison, 2015).

Three key points

There are three key points that contribute to understanding elasticity and stretching of the fascial form in yoga practice:

1. **Terminology:** What exactly is elasticity?
2. **Tensional integrity:** How does this work in the whole body?
3. **Energy storage capacity:** How is it optimized in practice?

Once these points are clarified, we can begin to see the foundation of **elasticity** as a valuable attribute of the whole tissue matrix. Then the advantages can be suitably derived from all variations of Hatha yoga styles (Ashtanga, Restorative, Vinyasa Flow, Kria, Iyengar, etc.).

Terminology

Elasticity is the capacity of a material to change shape (deform) under external force and resist internally, thereby returning to previous form (reform). This is measured as the difference

between the stiffness (resistance to deformation) and the elastic return (reformation).

A common misunderstanding in the classroom seems to be that (a) something has to be made of elastic to display elasticity and (b) when we let go of an elastic band or elasticated fabric, as it lies at rest, we are modeling our body at rest. Both of these ideas are inaccurate.

Stiffness is the resistance to deformation, but stiff springs (think of car parts) can have more elasticity than weaker springs (think of a slinky toy) because they store more energy and rebound more efficiently. Hard steel ball bearings bounce better than rubber balls (Levin, 2013). A material does not have to be made of elastic to demonstrate elasticity or have the capacity to store elastic energy.

We rest pre-tensioned, or pre-stiffened, by the architectural design of our bones and soft tissues, that is as a volume in space. Our elasticity does not just refer to the amount we can stretch. It refers to the ability or capacity to restore a change in shape. That is a suitable balance between stiffness (i.e. resistance to deformation) and elasticity (i.e. the ability to reform or restore the original shape). We are poised between the extremes of either state when we are relaxed. When we take up a yoga pose, to the extent that we can form the shape of the pose (deformation), it is counterbalanced by our ability to release it without imposition. If we force a stretch, then the ability of the tissues to restore (reformation) might be compromised.

The question is of a balance between these two states, rather than a focus on maximum flexibility for its own sake (Chapter 19). Appropriate counter poses, and respect for the elastic limit, invite an overall balance to the whole practice.

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When you pull on an elastic band or fabric you are measuring its stiffness (deformation capacity). When you release the band, the extent to which it immediately restores its original shape, is a measure of its elasticity (reformation ability). In fact our tissues rest pre-stiffened or pre-tensioned, poised for movement. If we go beyond the elastic limit, it is a different state: that of plasticity (Avison, 2015).

Plasticity: Beyond elasticity is a point where reformation is no longer possible. The so-called stretch is irreversible. The material does not spring back when you let it go and it retains the shape of the deformation. Between that point, the elastic limit, and the breaking or tearing point, is termed “plasticity”: when the new shape stays there. The subsequent changes are distinguished in terms of malleability and ductility or brittleness. All of these are properties of different types of materials.

Viscoelasticity is a characteristic of the living body: a solid-and-fluid medium.

“Viscoelasticity is a property of all tensegrity structures. It is a non-linear time dependent deformation. It is called ‘viscoelastic’ because when stressed it first behaves as if it were a liquid, then behaves as an elastic solid, not because it is made up from those two states. When stress is removed, it does it backwards and there is a splashdown (soft landing).”

Levin (2013).

These different qualities of stiffness to softness are different values or properties that, in combination within the tissues, make for a variety of tensional possibilities throughout the entire fascial forms in the overall architecture, whatever

the yoga position. Car springs have “dampers” on them to slow down the rate of elastic return (or reformation, see above). The fluids in our tissues act somewhat like dampers and give them so-called “viscoelastic properties”, which regulate the rate of resistance and reformation (Richards, 2012).

At every level of detail there is a balance of stiffness and elasticity in response to how we load or use them and how intelligently we accumulate useful adaptations, over those which may be less than optimal (such as forcing a stretch, for example, when the body is neither ready, nor able to counter it). Altogether, this exquisitely detailed matrix forms and develops the collagen matrix according to loading and the timing/frequency of that accumulated movement history. Habitual or deliberate strain patterns profoundly affect tissue morphology and mobility. Forced stretching can compromise the tissues’ innate ability to spring back if it isn’t counterbalanced, or sufficiently stiffened to optimize elasticity. This ability is expressed on the mat as a balance between appropriate stiffness, length and flexibility or compliance, for the individual.

Tensional integrity

One of the main features of our structural balance in motion is based upon the architectural principles of biotensegrity (Levin, 2012). This is the basis of our architecture. Unlike the elastic band resting on the table (untensioned), we remain tensioned all the time, to enclose and occupy space. By the basic laws of biotensegrity, every movement affects the whole and is transmitted through it.

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“Tensegrity structures are omni-directional, independent of gravity, load distributing and energy efficient, hierarchical and self-generating. They are also ubiquitous in nature, once you know what to look for.”

Levin (2009).

As biotensegrity (living tensegrity-based) structures, we have the ability to squeeze and draw in our structure and tissues as a whole, besides being able to stretch and fill or expand them, omni-directionally, for example, as in the filling and emptying of our bladders or lungs. The fascial fabric, bones, myofascia, organs, joints, vessels and cavities of our architecture are, paradoxically, held together and apart simultaneously, as volumes. They are globally contained through the tension/compression principles of biotensegrity architecture (Levin, 1990; Ingber, 1998; Flemons, 2006; Martin and Levin, 2012).

This is the basis of “primordial biologic structure” (Levin, 1990). Such a design is organized for stored energy capacity through the balance of stiffness to elasticity in the entire architecture on every scale. It is a foundational principle of human, indeed living, form, and one that we naturally explore in every yoga pose (asana) at the macrocosmic level of our whole form. Essentially, we seek to preserve and promote elastic integrity throughout the body.

Energy storage capacity

We constantly work under three phases of elastic range, not one. We can stretch and release, we can squeeze and release, and we can rest in a pre-tensioned state (neither squeezing nor releasing). The breath is the first place to recognize this law in motion. We breathe in and breathe out in the

inhale/exhale rhythm. However, we can then exhale more to empty or squeeze and release by inhaling. It is a three-phase process, with a pausing place in between, and one intent of yoga is to maintain this at a suitable balance. Dynamic sequences, slower restorative practice and the stillness of meditation are designed to explore this range and accumulate a resource of actively contained energy storage capacity throughout the body. This is enhanced by many of the yogic breathing practices of *pranayama*.

For the whole body, an overemphasis on softening, stiffening or stretching will not necessarily result in optimum elasticity in movement, whatever the activity (Chapter 8). The focus in yoga is on the refined transition and continuity along the spectrum either side of resting tension, to restore the phases of length-tension balance to the whole (see Figure 37.5).

While this may seem like an obvious statement shown in diagrammatic form, it is not always understood when applied practically. Even in yoga classes, breathing exercises are often performed without the distinctions of their value in preserving and promoting compliance and energy storage ability in the tissues. They have global implications of accumulating and fine-tuning elasticity throughout the fascial forms, in all aspects of the yoga practices, especially if the full range of the “breath mobius” is encouraged (Avison, 2015).

If we train the body in any one aspect, i.e. just squeezing (stiffening), just stretching or just softening (releasing), or in one direction or plane (i.e. just flexion), we can compromise overall elastic integrity. If we counter-pose and counteract to maintain that tensional balance, we frequently

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Figure 37.2

The tensional integrity of the whole body can be seen being used for balance. From nose tip to tail tip, this puppy reached out as part of a whole body-balancing act to drink from the pool, extending the tensional integrity to the tip of its tongue.

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load the tissues for optimum tensional integrity and range in all degrees of potential. Stretching becomes a feature of balance and suitable control, rather than a purpose for its own sake.

Instinctive stretching

This is a way of stretching that does not seem to compromise the tissues and which naturally regulates the dangers of overstretching globally. Animals can provide us with some clues.

Consider the cheetah, for example. Wild and domestic cats can rest and relax, stretch at a suitable time, move at great speed and pounce on prey. At rest, they become languid and serene. When they have rested, they frequently yawn-stretch their whole body to wake up their tissues after releasing them for a period of time. This particular type of stretching is called pandiculation (Bertolucci, 2011). It is what we call a “lengthening contraction” in yoga. It is a feature of movement with a purpose in nature, to reactivate the tissues after resting them and reorganize, or prime, the internal bonds and fluids ready for mobility (Chapter 9).

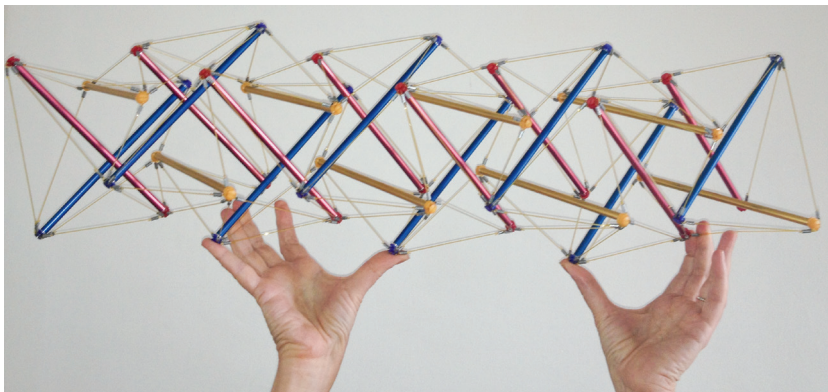


Figure 37.3

A tensegrity mast; originally presented by Kenneth Snelson as a system called “floating compression”, the term “tensegrity” was coined by Buckminster Fuller as “tensional integrity” and developed since by Dr Stephen Levin into biotensegrity (see Further information). It is an elastic architecture, balanced between stiffness and elasticity. It bounces and resists deformation, retaining its three-dimensional form.

Model by Bruce Hamilton (see Further information).

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When a cheetah anticipates prey (peak performance), the last thing they do is stretch. They prime their tissues by “squeezing” or stiffening them:



Figure 37.4

A classical model of the spine might suggest that it is a stacked column. This posture would not be possible if that were the case. Nor is it the muscular strength of Katie's standing foot that is holding her up. Her ability to hold and modify this pose is based upon the tensional integrity of her balance as a whole; or the “energy storage capacity”, which is the elasticity, perfectly poised in mutual tension/compression balance.

Reproduced with kind permission from Katie Courts (www.yoga-nutco.uk).

they stalk. They are ready to deploy the potential to pounce or sprint as required, maximizing their catapult capacity by tightening the global tissue matrix. Even their fur stands on end, perhaps for super-sensitivity to the task. Cheetahs, like many other mammals, focus and draw their tissues in to make them fit for purpose, globally. There is a time and a place, a dose and a degree, for stretching.

Research in British Columbia (see Further information) regarding how hibernating bears survive a winter of sleep without osteoporosis or degenerative conditions affecting their muscles after months of inertia, revealed an interesting instinctive habit. The bears get up around midday every day and perform 20–40 min of natural movement. They do gentle yoga-type movements, yawning-stretches in all directions, yawning and wriggling and pacing around, reanimating the tissues, before they settle back and sleep and hibernate for the next 24 h.

“Stretching is part of our nature, particularly when accompanied by yawning. This wholeheartedly reflects the ‘felt’ sense of tension and compression,

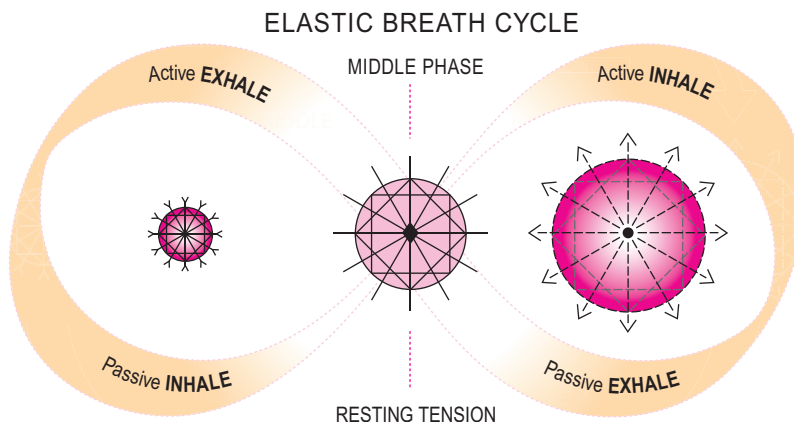


Figure 37.5

Resting tension is the middle phase, between the breaths. Inhalation is the stretch to expand and exhale restores resting tension. The “exhale plus” is the ability to squeeze and contain the breath, which restores resting tension by inhaling (Avison, 2014).

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Figure 37.6 Yawning stretch or pandiculation

A wild cheetah in up-down dog pose, its own version of Adho-Urdhva Muhka Svanasana. Notice the beautiful side bend of the cheetah in the background.

Reproduced with kind permission from Shane McDermott (www.wildearthilluminations.com).

or biotensegrity, in the system. It facilitates stretch-and-squeeze simultaneously and naturally self-regulates the tissues."

Bertolucci (2011).

How often do we remember to yawn-and-stretch after rest considering the time we spend in planes, trains and automobiles, not to mention sitting at our desks. Do we counterbalance them with yawning stretches? No cat, dog or bear would forget to do this after a period of inertia. It is instinctive to many animals. We, on the other hand, will force-stretch to reach the most extraordinary shapes as if that is the purpose of yoga.

Building a repertoire in the body of cumulative "enquiries" and "balances" is a clear advantage in improving vitality and may have distinct therapeutic value (Broad, 2012). It also speaks to issues of inertia that research shows to play a key role in many degenerative diseases (Henson et al., 2013). However, it is optimally used in a very

instinctive way, in order to explore those shapes for a given individual and maintain their value in enhancing true elasticity (recoil capacity). If someone is super-bendy or super-soft then they might find improved elasticity in tensioning, or stiffening, their network, rather than typically stretching it.

An example in practice: dog pose

Rather than forcing the stretch, time is taken to compress (stiffen) the body first, contain and gather it in (i.e. squeeze) to find the ground or base of support. This allows the heels, in the given example of Dog Pose (*Urdhva Muhka Svanasana*; Figure 37.7) to explore their tensional relationship throughout the back of the body, and invites the movement towards the ground, rather than forcing or reaching towards it, as if "stretching" is the goal. This, in turn, facilitates the knee to be opened naturally, without pushing it back or hyperextending the joint (Figure 37.7).

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This “listening approach” ensures that the fold at the hip does not pull the pelvis into a posterior tilt, which would, in turn, compromise the lumbar spine (lordosis), thus pulling on the shoulders or neck, and leaving Alexander straining to stretch to “achieve” the pose (Figure 37.8).

Balancing the tensional components includes the whole body at all its folding potentials (joints),



Figure 37.7

Once a principle ballet dancer, Alexander is more than capable of “stretching” his body in Dog Pose, heels to the ground. However, to optimize elasticity, every joint is seen for its role in the tensional balance of the whole. This includes the relationship of the back to the front and the continuity throughout their longitudinal organization, from fingertips to feet and coccyx to crown.

Model: Alexander Filmer-Lorch. Reproduced with kind permission from Charlie Carter Photography.

gathering in the continuous integrity of the tensional matrix. It naturally facilitates Alexander’s ability to squeeze/contain throughout the architecture and then “fill” the pose in 360 degrees, through an actively loaded stretch (Chapter 33) (or lengthening contraction), rather than strain for it through pulling on any particular aspect of the posterior tissues (Myers, 2009) (Chapter 35). Then he can explore the posture intelligently to his own elastic (or biotensegral) limit (Chapter 10) as a global gesture or expression to maintain a body-wide “biomotional integrity” (Avison, 2015). This is quite distinct from forcing the heel down or pushing the knee joint to “stretch into the pose” in a fight for length at all costs, which may be implied by stretching. This can cost natural elasticity (see earlier).

The key clue to this, or any other posture, is in the ease and flow of the breath and the subtle



Figure 37.8

Subtle adjustment gives feedback to tension the pose in suitable balance at each of the joints, breathing in to the potential length as it unfolds, without forcing stretch. Model: Alexander Filmer-Lorch.

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ability to transmit the breathing motion through the whole pose and body, rhythmically. That is through the expansion, release, squeeze, release cycle mentioned earlier, innate to a biotensegrity architecture (Chapter 11). When this is not compromised there is no need for extraneous, unnecessary tightness, floppiness or strain, only a suitable containment of the whole, to move or hold a pose, any yoga posture.

For the more dynamic practice, it encourages containment and thereby fostering spring-loaded tension or potential: the stored energy capacity for the next posture in a flowing sequence (Chapter 10). Thus it primes the tissues for effortless and contained transition between asanas at the variable speeds. We can, by fostering elasticity over stretching, nourish our loading history, through this conscious process of expansion and impansion of the global fascial matrix, whatever yoga forms we are seeking to morph into and practice. In any yogasana, we can adapt to the speed and transitions of differing styles. This invites us towards the yogic philosophy behind

seeking “*sthiram* and *sukham*”, the balance between the steady and the sweet if we invite the balance between tension and compression, in multidirectional (or omni-directional) wholeness. We accumulate balance in the elastic recoil potential and invite *sthiram/sukham* to the pose, as well as to the overall practice. We begin by accumulating balance, poise and adaptability, from speed to stillness and beyond.

Summary

Yoga practice is an exploration of balance and integration on many levels. Our fascial form rests in the neutral potential that we call “relaxed”, although the fascial tissues are biotensegrally orchestrated and always poised in tensional integrity for potential movement. As a force transmission system with innate elastic properties, dependent upon how we use the body and respect this unifying feature, the fascial matrix (and the forces it transmits) and yoga express and enhance each other in all their variety and possibility.

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Further information Bears at Grouse Mountain

At Grouse Mountain in Vancouver (<http://www.grousemountain.com/wildlife-refuge>) there is a BearCam facility and it is possible to watch Coola and Grinder in hibernation. Researchers and rangers in the park run a blog and films about the bears (and other wildlife at the reserve) and their hibernating habits.

Carter, C., photography.

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